Seasonal Analysis of Land Surface Temperature (LST) and Vegetation Cover Using NDVI

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1. INTRODUCTION

Urbanization significantly transforms natural landscapes by replacing vegetation and permeable surfaces with impervious materials such as concrete, asphalt, and industrial structures. These surfaces have higher heat absorption capacity and lower evapotranspiration rates compared to vegetated areas, leading to elevated surface temperatures. This phenomenon is widely recognized as the Urban Heat Island (UHI) effect.

Coastal cities like Kochi are particularly vulnerable to temperature changes due to:

- Rapid infrastructure development
- Port and industrial expansion
- Dense residential settlements
- Limited green cover

Unlike temperate regions, Kochi experiences a tropical monsoon climate (Köppen Am classification), characterized by high humidity and seasonal rainfall rather than extreme winter–summer contrasts. Seasonal temperature variation is influenced by:

- Solar radiation intensity
- Pre-monsoon heating
- Monsoon rainfall
- Post-monsoon surface moisture conditions

Land Surface Temperature (LST) derived from satellite thermal imagery provides valuable information about surface thermal behavior. NDVI, derived from reflectance bands, measures vegetation density and health.

When analyzed together, LST and NDVI help understand:

- Impact of built-up areas on heat accumulation
- Cooling effects of vegetation
- Seasonal thermal variation
- Urban heat island intensity

2. STUDY AREA

The study area is located in the Fort Kochi–Vallarpadam region of Ernakulam district, Kerala, India. The region includes:

- Vallarpadam International Container Transshipment Terminal
- Port infrastructure
- Residential settlements
- Backwaters and estuarine channels
- Limited vegetation patches

The area is characterized by mixed land use, with extensive industrial surfaces and water bodies. The proximity to coastal waters influences local microclimate conditions.

The spatial resolution of analysis is 30 meters, consistent with Landsat imagery.

3. DATA USED

Landsat 8/9 Collection 2 Level-2 data were obtained from the United States Geological Survey (USGS).

Dates analyzed:

- 01 December 2024 (Dry Season)
- 06 April 2025 (Peak Pre-Monsoon)
- 31 October 2025 (Post-Monsoon)
- 06 February 2026 (Pre-Monsoon Transition)

Bands Used:

- Band 10 (Thermal Infrared) → LST
- Band 4 (Red) → NDVI
- Band 5 (Near Infrared) → NDVI

4. METHODOLOGY

4.1 Land Surface Temperature Derivation

Thermal Band 10 was processed using raster calculations to convert digital values into temperature in degrees Celsius. Equal interval classification (5 classes) was applied for thematic mapping.

4.2 NDVI Calculation

NDVI was calculated using:

NDVI = (NIR − Red) / (NIR + Red)

NDVI values range from -1 to +1:

- < 0 → Water
- 0–0.2 → Built-up / Bare soil
- 0.2–0.4 → Sparse vegetation
- 0.4 → Moderate to dense vegetation

Red-to-green color scheme was applied for visualization

5. RESULTS AND ANALYSIS

5.1 Seasonal Variation of Land Surface Temperature (LST)

The seasonal analysis of LST reveals noticeable variation in surface temperature across the study area. Although Kochi experiences a tropical monsoon climate with relatively moderate annual temperature variation, the spatial distribution of heat differs significantly due to land use characteristics.

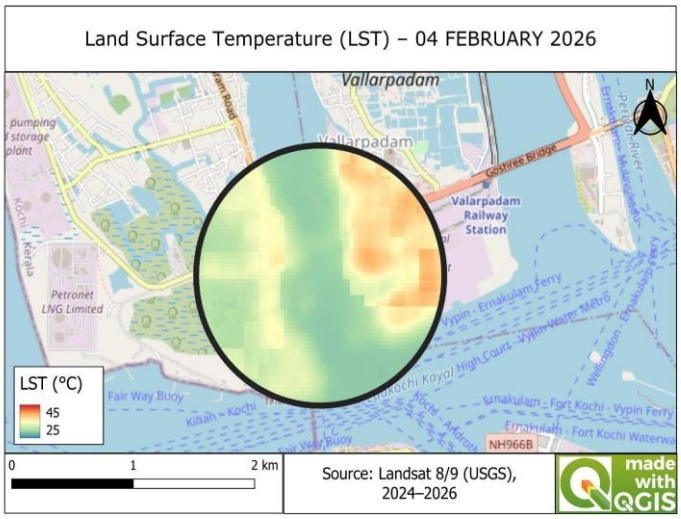
5.1.1 February 2026 – Pre-Monsoon

- Minimum LST = **28.79 °C**
- Maximum LST = **39.98 °C**
- Mean LST = **32.66 °C**
- Standard Deviation = 3.04

February represents the beginning of pre-monsoon warming. Interestingly, the mean temperature (32.66 °C) is slightly lower than December's mean (33.35 °C). This may be due to differences in atmospheric conditions on the satellite acquisition date.

Unlike December, the minimum temperature in February is much higher (28.79 °C), indicating reduced cooling contrast. The lower standard deviation (3.04) suggests more uniform surface heating.

This period shows gradual thermal build-up before peak summer.



5.1.2 October 2025 – Post-Monsoon

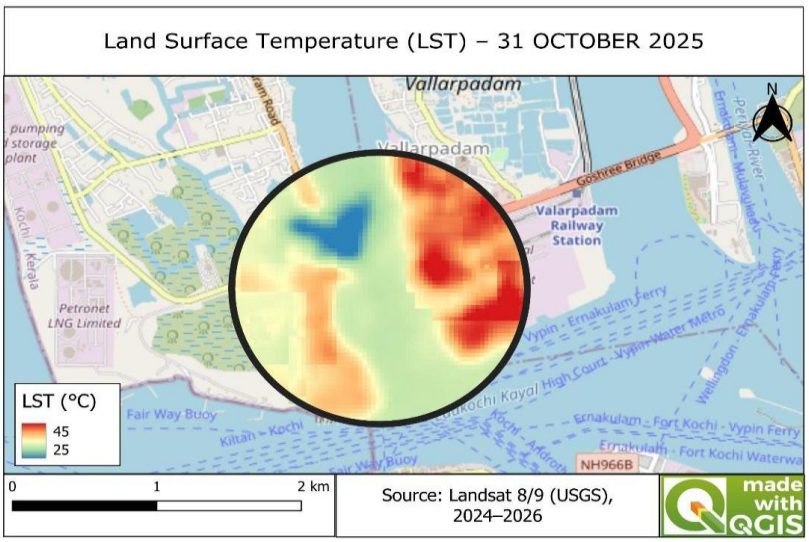
- Minimum LST = **22.21 °C**
- Maximum LST = **47.27 °C**
- Mean LST = **35.66 °C**
- Standard Deviation = 4.79

Post-monsoon conditions show moderate mean temperatures (35.66 °C), lower than April but higher than December.

Interestingly, the maximum temperature (47.27 °C) is slightly higher than April's maximum. This suggests localized extreme heating over industrial surfaces, possibly due to:

- Clear skies after monsoon
- Residual heat storage in port infrastructure
- Surface drying after rainfall

The higher standard deviation (4.79) indicates renewed spatial contrast in temperature distribution.



5.1.3 April 2025 – Summer

- Minimum LST = **28.19 °C**
- Maximum LST = **46.61 °C**
- Mean LST = **37.32 °C**
- Standard Deviation = 3.58

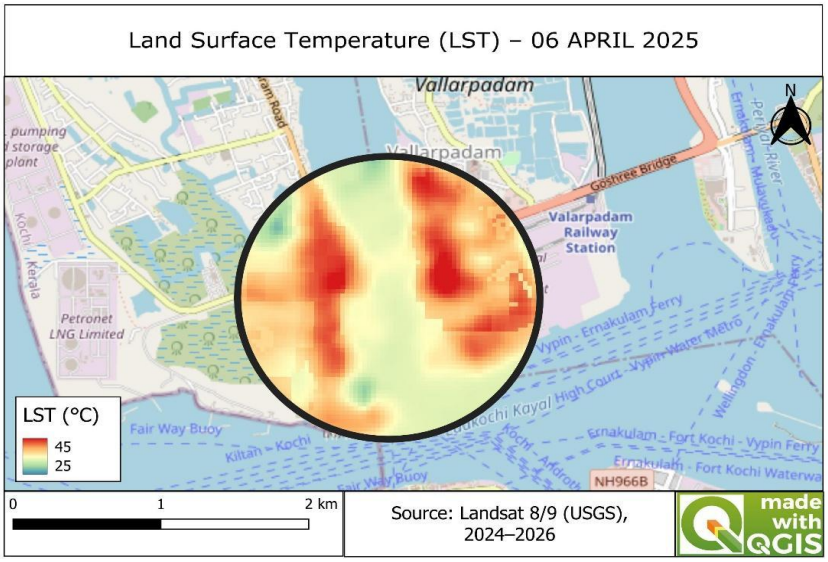
April exhibits the highest mean temperature among all seasons analyzed. The mean LST of 37.32 °C clearly indicates peak thermal stress in the region.

Maximum temperatures exceed 46 °C over built-up and industrial zones. The intense heating during April can be attributed to:

- High solar radiation
- Low soil moisture
- Clear sky conditions
- Reduced cloud cover

Water bodies continue to maintain comparatively lower temperatures, reinforcing their role as natural cooling elements.

This period clearly demonstrates the strongest Urban Heat Island effect.



5.1.4 December 2024 – Dry Season

- Minimum LST = **18.13 °C**
- Maximum LST = **44.72 °C**
- Mean LST = **33.35 °C**
- Standard Deviation = 5.15

December represents the dry season in Kochi. The minimum temperature of 18.13 °C is primarily associated with water bodies and shaded zones. These areas show strong cooling due to high heat capacity and evaporative processes.

However, despite being a relatively cooler season, maximum temperatures reach 44.72 °C over built-up port and industrial surfaces. This indicates significant heat retention by impervious materials such as concrete and asphalt.

The relatively high standard deviation (5.15) suggests strong spatial temperature contrast between water bodies and built-up areas.

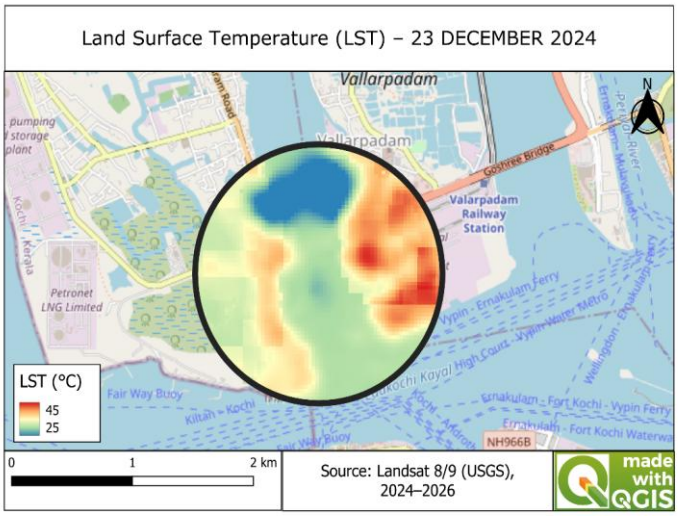
Spatially, the hottest zones are concentrated over:

- Industrial surfaces
- Road infrastructure

Cooler zones are observed over:

- Backwaters

This confirms the influence of land use on surface temperature distribution.



5.2 Comparative Seasonal Analysis

When comparing mean LST values:

- April 2025 → **37.32 °C (Highest)**
- October 2025 → **35.66 °C**
- December 2024 → **33.35 °C**
- February 2026 → **32.66 °C**

This confirms that peak pre-monsoon summer (April) experiences maximum overall surface heating.

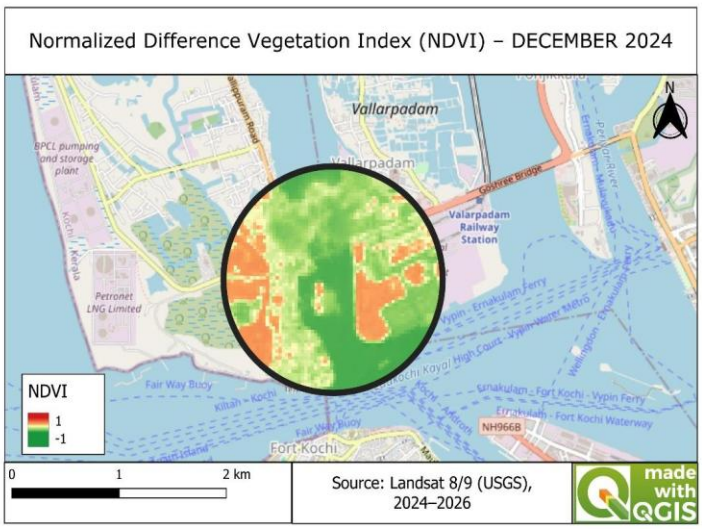
The variation between minimum and maximum temperatures in December and October is larger than in February and April, indicating greater spatial heterogeneity during those periods.

5.3 NDVI Analysis

December 2024

- Minimum NDVI = -0.070
- Maximum NDVI = 0.485
- Mean NDVI = 0.148

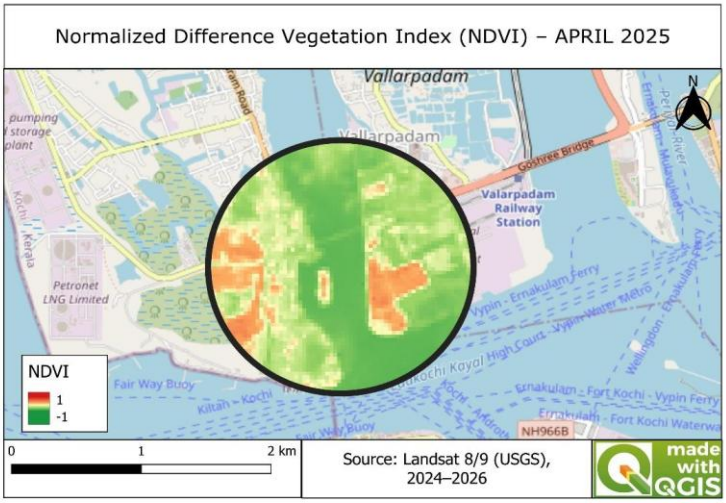
The low mean NDVI (0.148) indicates sparse vegetation across the study area. Large portions of the region fall under low NDVI classes (0–0.2), representing built-up areas and bare surfaces.



April 2025

- Minimum NDVI = -0.130
- Maximum NDVI = 0.731
- Mean NDVI = 0.221

April shows a slightly higher mean NDVI (0.221), suggesting moderate vegetation activity in certain patches. However, vegetation remains fragmented and limited compared to built-up surfaces.



5.4 Relationship Between LST and NDVI

A clear inverse relationship between NDVI and LST is observed:

- Areas with NDVI < 0.2 consistently show higher LST values.
- Vegetated areas with NDVI > 0.4 show comparatively lower surface temperatures.
- Water bodies show negative NDVI values and lowest temperatures.

During April 2025, zones with NDVI below 0.2 correspond to temperatures above 40 °C. This strongly supports the cooling role of vegetation through evapotranspiration and shading.

The dominance of port infrastructure and industrial surfaces contributes to persistent high-temperature zones across seasons.

6. Environmental Implications and Recommendations

6.1 Environmental Implications

The seasonal analysis of Land Surface Temperature (LST) and NDVI in the Fort Kochi–Vallarpadam region reveals important environmental implications related to urban development and land use.

The study shows that built-up and industrial surfaces consistently exhibit higher surface temperatures across all seasons. During peak pre-monsoon (April 2025), the mean LST reached 37.32 °C, with localized hotspots exceeding 46 °C. Even during relatively cooler periods such as December 2024, maximum temperatures exceeded 44 °C in industrial zones.

This indicates the presence of localized Urban Heat Island (UHI) effects, particularly over:

- Port infrastructure
- Container terminals
- Road networks
- Industrial surfaces

The NDVI results further confirm that vegetation cover in the study area is limited. The mean NDVI values (0.148 in December and 0.221 in April) suggest sparse and fragmented vegetation distribution. Areas with low NDVI values correspond spatially to zones with high LST values.

This inverse relationship highlights that vegetation plays a significant role in regulating surface temperature through:

- Shading
- Evapotranspiration
- Reduction of surface heat absorption

Water bodies within the study area consistently showed the lowest surface temperatures, demonstrating their natural cooling effect in coastal environments.

If urban expansion continues without increasing green cover, surface heating may intensify, potentially affecting:

- Local microclimate
- Human thermal comfort
- Energy demand for cooling

6.2 Recommendations

Based on the findings of this study, the following recommendations are suggested:

1. Increase Urban Green Cover

Planting trees and developing green buffer zones around port and industrial areas can significantly reduce surface temperatures. Urban vegetation enhances evapotranspiration and reduces heat accumulation.

2. Develop Green Belts Around Industrial Infrastructure

The Vallarpadam container terminal and surrounding industrial zones would benefit from structured green belts to mitigate localized heat buildup.

3. Promote Heat-Reflective and Sustainable Materials

Use of reflective roofing materials and permeable surfaces can reduce heat absorption in built-up areas.

4. Protect and Maintain Water Bodies

Backwaters and estuarine systems play a crucial role in regulating local temperature. Their preservation is essential for maintaining environmental balance.

5. Integrate Thermal Analysis in Urban Planning

Urban development projects in coastal cities like Kochi should incorporate geospatial thermal assessments to ensure climate-resilient planning.

7. CONCLUSION

This study analyzed seasonal variation in Land Surface Temperature (LST) and vegetation distribution using NDVI in the Fort Kochi–Vallarpadam region of Kerala. Using Landsat 8/9 satellite data, LST was derived for four seasonal periods, and NDVI was calculated for two representative seasons.

The results reveal clear seasonal variation in surface temperature. The highest mean LST was recorded during the peak pre-monsoon season (April 2025) with a mean value of 37.32 °C, while comparatively lower mean temperatures were observed during the dry season (December 2024) and pre-monsoon transition period (February 2026). Post-monsoon conditions (October 2025) showed moderate mean temperatures but localized extreme hotspots.

The spatial distribution of temperature strongly reflects land use characteristics. Built-up and industrial port infrastructure consistently exhibited higher surface temperatures, whereas water bodies demonstrated the lowest temperatures across all seasons. This confirms the significant influence of surface materials on thermal behavior.

NDVI analysis indicates sparse and fragmented vegetation cover in the study area, with mean NDVI values of 0.148 (December 2024) and 0.221 (April 2025). A clear inverse relationship between NDVI and LST was observed: areas with low vegetation cover corresponded to higher surface temperatures, while vegetated patches and water bodies exhibited cooling effects.

The findings confirm the presence of localized Urban Heat Island effects in the Vallarpadam port region. The dominance of impervious surfaces and limited green cover contributes to increased thermal stress, especially during the pre-monsoon summer season.

Overall, this study demonstrates the effectiveness of integrating LST and NDVI analysis using remote sensing techniques to understand seasonal thermal dynamics in coastal urban environments. The results highlight the importance of increasing vegetation cover and incorporating thermal considerations into urban planning strategies to promote climate resilience and environmental sustainability.